

CE 310 — Week 4 In-Class Exercise: When Rare Becomes Routine

Dataset: NOAA Daily Precipitation — Tucson International Airport, 2013–2022 **Topic:** Poisson Distribution · Event Rate (λ) · PMF · CDF · Return Periods

Before You Begin

Files needed: - Week4_Tucson_Precipitation_Daily_2013_2022.csv — 3,652 daily rows of precipitation data - Wk04.04_In_Class_Answer_Sheet.xlsx — download from D2L and enter all answers here

Dataset note: This is real station data from NOAA's Climate Data Online for Tucson International Airport (Station USW00023160). Each row is one calendar day, 2013–2022. The column PRCP_in records daily total precipitation in inches; IsRainDay = 1 if any precipitation was recorded, 0 otherwise.

Answer Sheet instructions: Type every numerical answer in Col E of the row matching the question label — A1 in row A1, B3 in row B3. For ★ questions, also enter the intermediate value in Col D. For Col C, type the Excel formula or method you used.

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761)'. Without the apostrophe Excel evaluates it and your method is lost. **Screenshots are not accepted;** the typed formula is what earns the approach credit.

Background

This week you'll fit λ from real precipitation data and use it to evaluate exactly how rare a “rare” event really is.

Excel Quick Reference

Goal	Formula
Count non-empty cells to verify row count	=COUNTA(A:A)-1 — subtract 1 to exclude the header
Count events in a category	=COUNTIFS(E2:E3653, "Summer", G2:G3653, 1)
Poisson PMF — $P(X = k)$	=POISSON.DIST(k, lambda, FALSE)
Poisson CDF — $P(X \leq k)$	=POISSON.DIST(k, lambda, TRUE)
$P(X \geq k)$ — at least k events	=1 - POISSON.DIST(k-1, lambda, TRUE)
$P(X > k)$ — more than k events	=1 - POISSON.DIST(k, lambda, TRUE)
Return period formula	=1 - EXP(-N/T) where N = years, T = return period
Build PMF table	Fill down: =POISSON.DIST(A2, \$B\$2, FALSE) where A2 = k value

Setup (~10 min)

You do all of this week's work in **one file: your Answer Sheet**. The data, your scratch work, and your charts all live there, so everything you built is in the file you hand in. Add as many sheets as you like — only Instructions, Answers, Work, Charts and Written have to keep their names.

1. Import **Week4_Tucson_Precipitation_Daily_2013_2022.csv** onto a sheet named Data that you add to your Answer Sheet:
 - Data → Get Data → From Text/CSV → Load.
2. Confirm **3,652 rows** of data: $=\text{COUNTA}(A:A)-1$
3. Confirm the **7 column names** in row 1:
 - Date, Year, Month, Day, Season, PRCP_in, IsRainDay
 - **Season** = "Winter" (DJF), "Spring" (MAM), "Summer" (JJA), "Fall" (SON)
 - **IsRainDay** = 1 if PRCP_in > 0; = 0 otherwise
 - **Month** = 1–12 (integer)

Why IsRainDay? A rain day is an event in the Poisson sense — it either happens or it doesn't on a given day. Counting rain days per month gives us k (the observed event count). Averaging those counts across months gives us λ (the estimated rate).

Poisson validity check (ungraded): Before applying POISSON.DIST, verify the rain-day data meets the three Poisson conditions: (1) discrete events — each day is either a rain day or not; (2) independence — whether it rained yesterday does not change today's probability; (3) constant rate — we assume λ is stable across the 2013–2022 period. Condition 3 is the one most worth scrutinizing: your seasonal analysis in Sections A and B will reveal whether a single λ is adequate, or whether seasonal rates are needed instead.

Section A — Marginal Event Rate (~25 min)

Section A fits a single event rate to the entire ten-year record and asks what the Poisson model predicts from it. One λ , one distribution, no seasons.

All Section A answers use COUNTIF, COUNTIFS, or POISSON.DIST. Round percentages to **3 decimal places**.

A1 (2 pts) — Total Rain Days

How many days in 2013–2022 recorded any precipitation (IsRainDay = 1)?

A2 (2 pts) — Overall Lambda

The dataset covers exactly **120 months** (10 years × 12 months). Compute the overall average rain days per month:

$$\lambda_{\text{overall}} = \text{A1 value} / 120$$

λ is the single most important number in a Poisson model. It is both the mean and the variance of the distribution. A system designed for the average (λ) will fail 50% of the time — because half of all months will exceed the mean.

A3 (2 pts) — P(Dry Month): PMF at $k = 0$

What fraction of months have **zero** rain days, if rain follows a Poisson distribution with your λ_{overall} ?

Multiply by 100 to express as a percentage.

`POISSON.DIST(k, lambda, FALSE)` computes the **probability mass function** — the probability of exactly k events. FALSE selects the PMF; TRUE selects the CDF.

A4 (2 pts) — P(Exactly 5 Rain Days): PMF at $k = 5$

A5 (2 pts) — P(At Most 5 Rain Days): CDF at $k = 5$

TRUE gives the **cumulative distribution function** — $P(X \leq k)$. Notice how $A5 > A4$: the CDF accumulates all probabilities from $k = 0$ up through $k = 5$.

A6 (2 pts) — Summer Rain Days

How many of the 3,652 days are in Summer (Season = "Summer") AND are rain days (IsRainDay = 1)?

A7 (2 pts) — Summer Lambda

The dataset has **30 Summer months** (10 years $\times$ 3 summer months per year). Compute:

$$\lambda_{\text{summer}} = A6 \text{ value} / 30$$

A8 (2 pts) — P(Exactly 10 Rain Days in a Summer Month)

Compare the Summer λ you computed in A7 to the overall λ from A2. The ratio shows how much more rain-active summer is. This is the same conditioning insight from Week 3 — now formalized as a rate.

Section B — Seasonal Poisson Analysis

Section B breaks that single rate apart by season. One annual λ hides a monsoon, and a system designed to the average rate is designed for a month that never happens.

Report percentages to 3 decimal places, λ values to 2 decimal places.

B1 (4 pts) — Spring Rain Days

How many Spring days (Season = "Spring") are rain days?

B2 ★ (5 pts) — Spring Lambda

The dataset has 30 Spring months.

Col D: number of Spring months in the dataset · **Col E:** $\lambda_{\text{spring}} = B1 / 30$

B3 (4 pts) — P(Dry Spring Month)

Compare to A3 (dry month at λ_{overall}). How does the probability of a completely dry month change in spring?

B4 (4 pts) — P(Unusually Wet Spring): $P(X \geq 5)$

An “unusually wet” spring month is one with 5 or more rain days. At the spring rate:

The complement rule: $P(X \geq 5) = 1 - P(X \leq 4)$. This is how you compute “exceedance probabilities” from a Poisson CDF.

B5 (4 pts) — July Rain Days

How many days in July (Month = 7) are rain days?

B6 ★ (5 pts) — July Lambda

July λ is computed per year (10 Julys in the dataset), not per month.

Col D: number of complete Julys in the dataset · **Col E:** $\lambda_{\text{july}} = B5 / 10$

July λ should be dramatically higher than spring λ . This is Tucson’s North American Monsoon. Warm moist air from the Gulf of California drives intense convective storms in July and August — a phenomenon so reliable it has its own name and its own engineering design requirements.

Engineering Context: ASCE 7-22 specifies design loads using return periods — the average time between exceedance events. Under the Poisson model:

$$P(\geq 1 \text{ event in } N \text{ years}) = 1 - e^{(-N/T)} = 1 - e^{(-10/25)} = 32.97\%$$

B7 (4 pts) — P(Dry July): The Monsoon Test

Your answer will be a very small percentage — report it to **3 decimal places** (for example, 0.123), and do **not** round it down to 0. A dry July in Tucson is nearly impossible — the monsoon produces rain on 10+ days in a typical July — but “nearly impossible” is still a number, and reporting it as zero is the mistake this question is about. An engineer who designs outdoor drainage assuming “it might not rain” in July is designing for the wrong distribution.

B8 (4 pts) — P(Extreme July): $P(X > 15)$

An extreme July has more than 15 rain days. At λ_{july} :

ArcE Framing: ASHRAE 90.1-2022 requires roof drainage systems sized for design storm intensity — not the annual average. For Tucson, the 100-year, 1-hour intensity is approximately 2.5 in/hr (NOAA Atlas 14).

B9 (4 pts) — Winter Rain Days

How many Winter days (Season = “Winter”) are rain days?

B10 ★ (5 pts) — Winter Lambda

Col D: number of Winter months in the dataset · **Col E:** $\lambda_{\text{winter}} = B9 / 30$

B11 (4 pts) — P(Dry Winter Month)

Compare B11 (winter) to B3 (spring) and A3 (overall). Which season has the highest probability of a completely dry month? Explain physically why.

B12 (4 pts) — P(Wet Winter Month): $P(X \geq 8)$ **B13 (4 pts) — Return Period Calculation**

A stormwater system is designed for the **25-year storm** (return period $T = 25$ years). What is the probability that this design storm occurs **at least once** during a **10-year** project analysis period?

Using the Poisson return-period formula:

$$P(\text{at least 1 in } N \text{ years}) = 1 - e^{(-N/T)}$$

Important: Use `EXP()` for this formula — **not** `POISSON.DIST()`. The approach column requires `EXP`.

Why `EXP` and not `POISSON.DIST`? This is $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{(-\lambda)}$, where $\lambda = N/T = 10/25 = 0.4$ events expected in 10 years. Writing it with `EXP` makes the exponential structure explicit and matches how engineers write return-period calculations in practice. The result should surprise you — a 25-year storm is far from rare over a 10-year design life.

B14 ★ (5 pts) — Is an Above-Average July Common?

The July mean is $\lambda_{\text{july}} = 10.3$ rain days. You might expect that exactly half of all Julys are above average. Is that true?

Compute $P(X > 10 \mid \lambda_{\text{july}})$ — the probability that a randomly chosen July has **more** rain days than the mean:

Col D: $P(X \leq 10 \mid \lambda_{\text{july}})$ · **Col E:** $P(X > 10 \mid \lambda_{\text{july}})$

Since λ_{july} is not a whole number (10.3), the Poisson distribution is very slightly right-skewed — $P(\text{above mean})$ is slightly less than 50%. This asymmetry grows larger at small λ and shrinks toward 50% as $\lambda \rightarrow \infty$.

Where the working goes. Your Answer Sheet has a **Work** sheet with a labelled, pre-filled block for each table below. Build them there and nowhere else — the charts are scored by looking at the cells they plot, so a table in a blank corner of another sheet is a table your preceptor has to hunt for.

Required Charts (15 pts)

Both charts belong on the **Charts sheet** of your Answer Sheet, in the outlined boxes. That sheet lists what each chart is graded on.

Build the chart wherever your data is, then right-click it -> **Move Chart -> Object in -> Charts**, and drag it into the matching box. Leave it as a live chart — do not replace it with a picture of itself. Your preceptor clicks the chart to see which cells it plots, which is how a correct chart built from the wrong column gets caught.

Chart 1 (7 pts) — PMF Comparison: Spring vs. Summer

The Work sheet of your Answer Sheet has this table ready in the block headed *Chart 1* — $k = 0$ through 15 already down column A, at rows 15 to 30. Fill the two yellow columns:

B15: =POISSON.DIST(\$A15, \$B\$8, FALSE) (spring)

C15: =POISSON.DIST(\$A15, \$B\$9, FALSE) (summer)

Fill both down to row 30, then plot A15:C30 as a clustered column chart. B8 and B9 are your spring and summer λ from the seasonal block just above.

The one mistake that ruins this chart: forgetting the \$ signs on lambda. Write \$B\$8, not B8. Without them the λ reference walks down the sheet as you fill, so every bar uses a different λ and the chart is garbage. Let the k cell shift; lock λ .

- Title: Poisson PMF — Spring vs. Summer, Tucson 2013–2022
- Axes: Rain Days per Month (k) on x, Probability on y (formatted as %)
- Two distinct colors with a legend

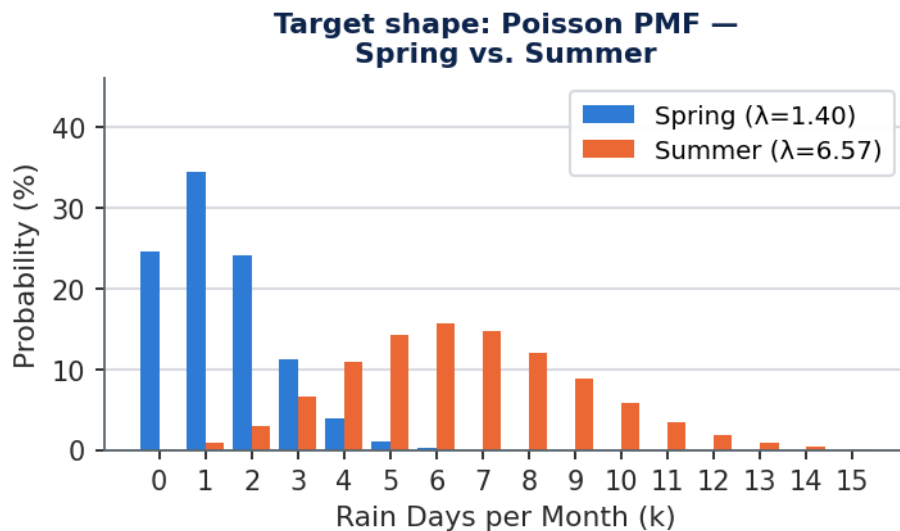


Figure 1: Target shape for Chart 1

Chart 2 (8 pts) — Lambda by Season Profile

The Work sheet has the four seasons already listed in meteorological order in the block headed *Chart 2*, at rows 7 to 10. Type your λ values into B7 to B10 — Winter, Spring and Summer are answers you already have; Fall is the one you compute here, COUNTIFS(Season = "Fall",

IsRainDay = 1) / 30. Plot A7:B10 as a clustered column. Because the order is given, the *meteorological order* point comes free.

- Title: Mean Rain Days per Month by Season – Tucson 2013–2022
- Y-axis: λ (rain days / month)
- Data labels on each bar showing the λ value

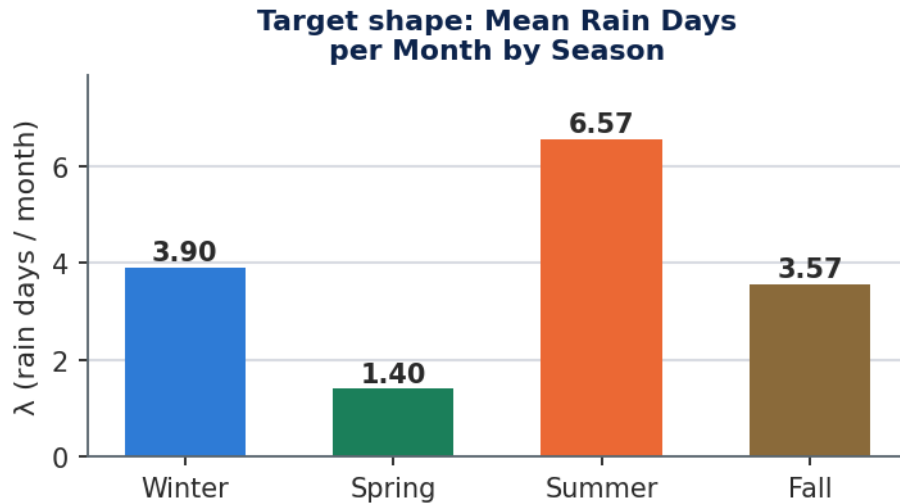


Figure 2: Target shape for Chart 2

Memo Deliverable (9 pts)

Type your memo directly in the **Written sheet** of your Answer Sheet Excel file. The sheet has labeled yellow cells for each paragraph (P1, P2, P3). Press Alt+Enter within a cell to add new lines. No separate PDF file is needed.

To: CE 310 Instructor
 From: [Your Name], [Your Major]
 Date: [Date]
 Subject: Week 4 In-Class Exercise – Poisson Analysis of Tucson Precipitation, 2013–2022

Paragraph 1 (3 pts) — The Event Rate: State your overall λ , summer λ , and spring λ . Using the PMF results from Section A and B, describe what the Poisson model says about the likelihood of a “dry month” (zero rain days) in each season. Cite at least two specific PMF values.

Paragraph 2 (3 pts) — Exceedance and Design: Using your B8, B12, and B13 results, explain the concept of exceedance probability and its relationship to return periods. In your own words, explain why $P(X > 15 \text{ in July})$ and $P(25\text{-year storm in 10 years})$ are both Poisson questions, and what they have in common methodologically.

Paragraph 3 (3 pts) — Monsoon drainage design and building envelope:

A building envelope consultant is specifying roof drainage and window flashing for a medium office building in Tucson. Using your seasonal λ values, explain why the monsoon season (Summer/July) dominates the drainage design requirement while winter rainfall drives a different set of concerns

(slow, sustained events vs. intense convective bursts). Reference the ASHRAE 90.1 design-storm context from the ArcE Framing callout. Cite at least one Poisson probability from Section B.

Submission: Save as CE310_W4_<NetID>.xlsx and upload to the Week 4 D2L dropbox. Enter your NetID in the **NetID** cell at the top of the Answers sheet — that cell is what matches your work to your student record. A 5-point deduction applies if it is blank.

Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet — a blank NetID cell takes a 5-point deduction
 - ☐ File saved as .xlsx and named CE310_W4_<NetID>.xlsx
 - ☐ Sheet tabs not renamed, reordered, or deleted
 - ☐ Every answer typed into its designated cell in **Col E** — answers placed anywhere else receive no credit
 - ☐ Your formula typed as text in **Col C**, with an apostrophe before the = — screenshots are not accepted
 - ☐ ★ questions also have their intermediate value in **Col D**
 - ☐ Both charts sitting in their boxes on the Charts sheet, as live charts and not pictures, with titles and axis labels
 - ☐ Memo written in the Written sheet
 - ☐ Uploaded to D2L by **9:00 AM next Wednesday**
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Grading Summary

Component	Points
Section A — A1–A8	16
Section B — B1–B14	60
Chart 1 — PMF comparison, Spring vs. Summer	7
Chart 2 — Seasonal lambda profile	8
Memo (3 paragraphs)	9
Total	100